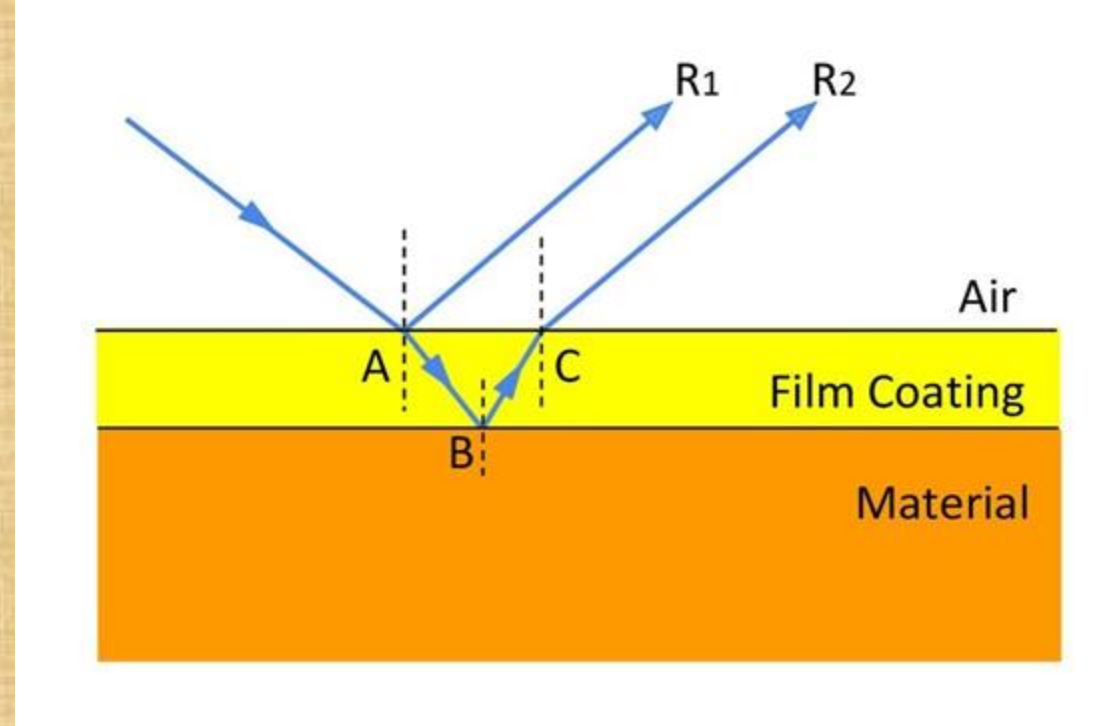


ANTI-REFLECTION COATING

4 times for 3 marks



Amplitude of ray 1 = Amplitude of ray 2

$$\left(\frac{\mu_f + \mu_a}{\mu_f - \mu_a}\right)^2 = \left(\frac{\mu_g + \mu_f}{\mu_g - \mu_f}\right)^2$$

Take square root on both sides

$$\left(\frac{\mu_f + \mu_a}{\mu_f - \mu_a}\right) = \left(\frac{\mu_g + \mu_f}{\mu_g - \mu_f}\right)$$

$$(\mu_f + \mu_a)(\mu_g - \mu_f) = (\mu_g + \mu_f)(\mu_f - \mu_a)$$

Taking $\mu_a = 1$

$$(\mu_f + 1)(\mu_g - \mu_f) = (\mu_g + \mu_f)(\mu_f - 1)$$

$$\mu_f(\mu_g - \mu_f) + \mu_g - \mu_f = \mu_f(\mu_g + \mu_f) - \mu_g - \mu_f$$

$$\mu_f\mu_g - \mu_f^2 + \mu_g - \mu_f = \mu_f\mu_g + \mu_f^2 - \mu_g - \mu_f$$

$$2\mu_g = 2\mu_f^2$$

$$\mu_f = \sqrt{\mu_g}$$

REFRACTIVE INDEX OF A LIQUID

3 times for 4 marks

$$slope_{air} = \frac{4R\lambda}{\mu_{air}} = \left[\frac{D_{n+m}^2 - D_n^2}{m} \right]_{air}$$



$$slope_{liq} = \frac{4R\lambda}{\mu_{liq}} = \left[\frac{D_{n+m}^2 - D_n^2}{m} \right]_{liq}$$



$$\frac{\mu_{liq}}{\mu_{air}} = \frac{slope_{air}}{slope_{liq}} = \frac{[D_{n+m}^2 - D_n^2]_{air}}{[D_{n+m}^2 - D_n^2]_{liq}}$$

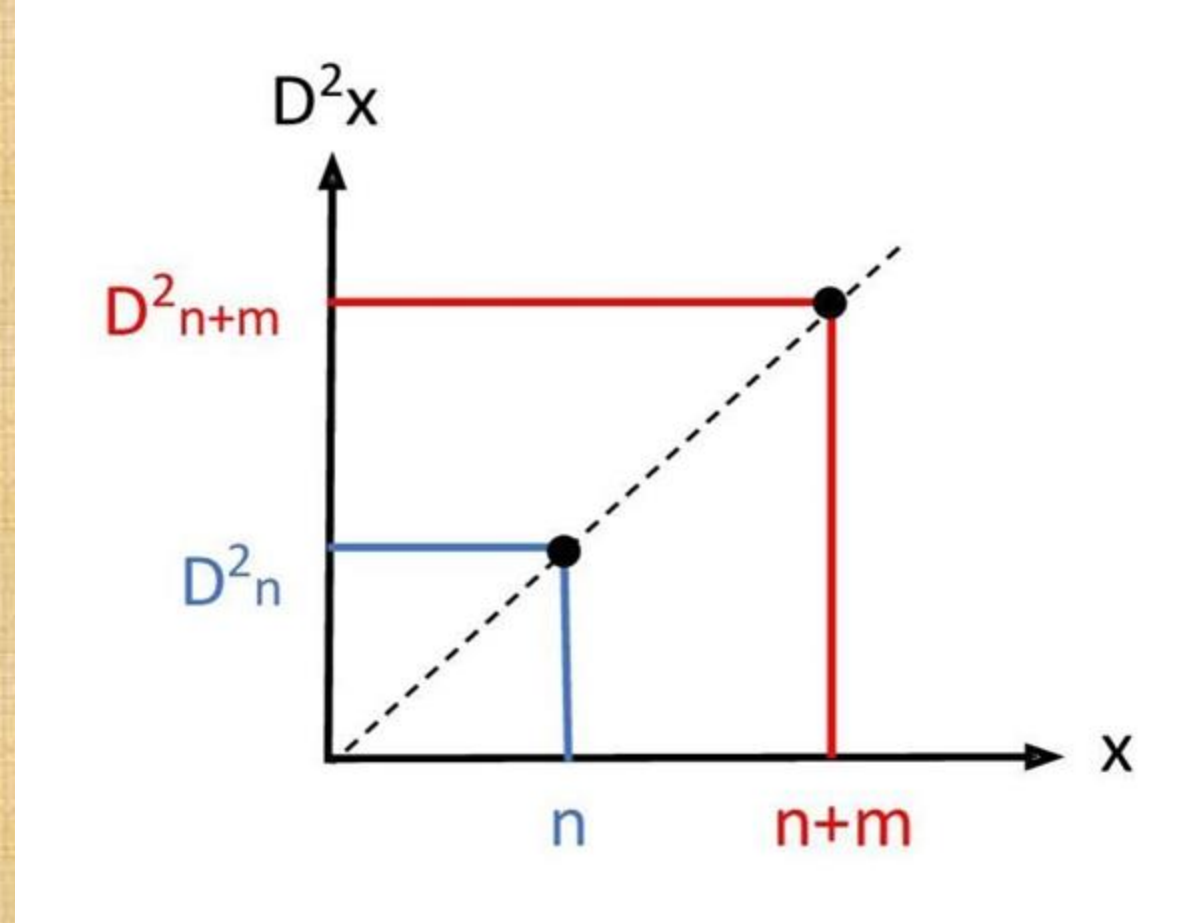
Since μ_{air} is 1



$$\mu_{liq} = \frac{slope_{air}}{slope_{liq}} = \frac{[D_{n+m}^2 - D_n^2]_{air}}{[D_{n+m}^2 - D_n^2]_{liq}}$$

WAVELENGTH OF LIGHT: NEWTON'S RINGS

4 times for 3 marks



$$slope = \frac{y_2 - y_1}{x_2 - x_1} = \frac{D_{n+m}^2 - D_n^2}{n + m - n} = \frac{D_{n+m}^2 - D_n^2}{m}$$

$$r_n^2 = \frac{nR\lambda}{\mu}$$

Radius of the dark ring is related to square root of the ring number

$$D_n^2 = \frac{4nR\lambda}{\mu}$$

For diameter $D = 2r$

$$D_{n+m}^2 = \frac{4(n+m)R\lambda}{\mu}$$

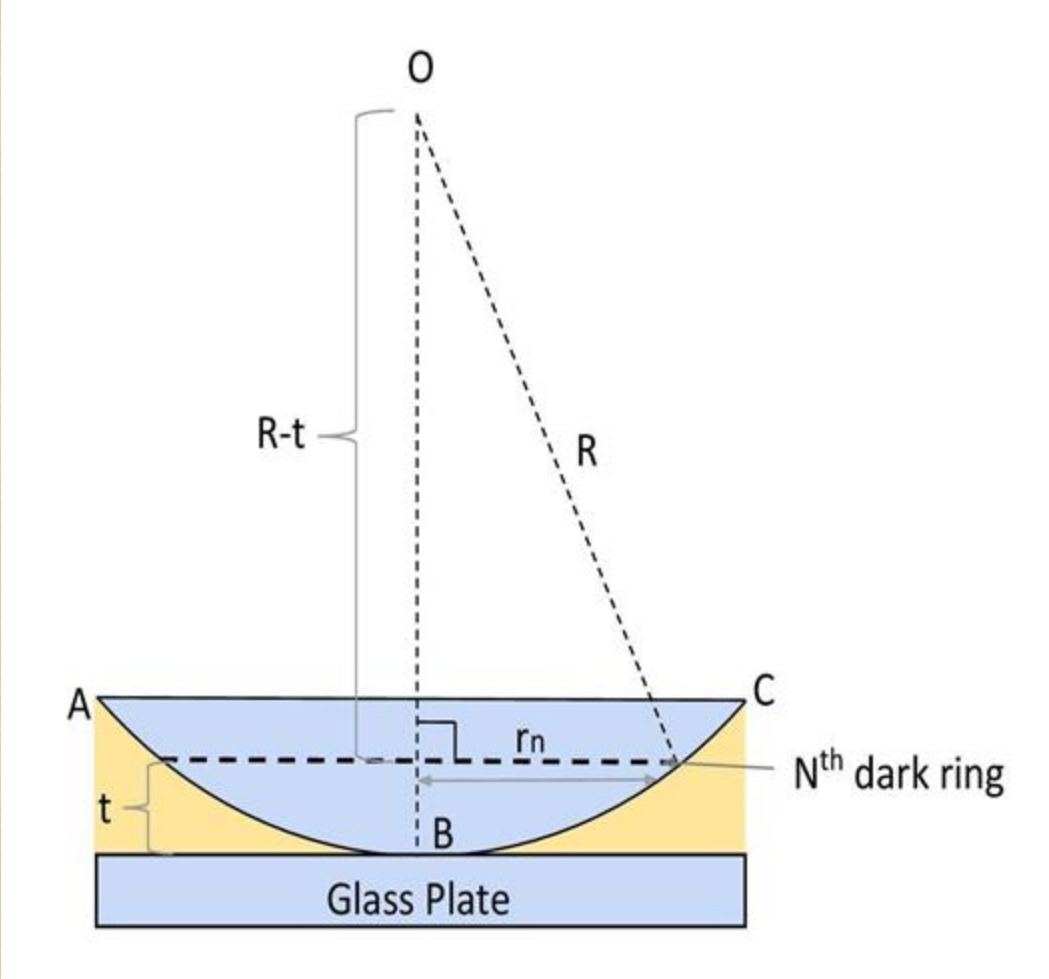
$$slope = \frac{D_{n+m}^2 - D_n^2}{m} = \frac{4nR\lambda}{\mu}$$

for air film $\mu = 1$

$$\lambda = \frac{D_{n+m}^2 - D_n^2}{m} = \frac{slope}{4R}$$

NEWTON'S RINGS NTH DARK RING AND RING NUMBER

2 times for 5 marks



Pythagoras theorem:

$$R^2 = R^2 + t^2 - 2R + r_n^2$$

't' (of the order of 10^{-7}) is very small,
so, t^2 even smaller, hence neglected

$$2Rt = r_n^2$$

$$2t = \frac{r_n^2}{R}$$

$$2\mu t \cos r = n\lambda$$

The interference condition for
dark band in reflected system

For normal incidence $r \sim 0$ and
hence $\cos r \sim 1$

$$2\mu t = n\lambda$$

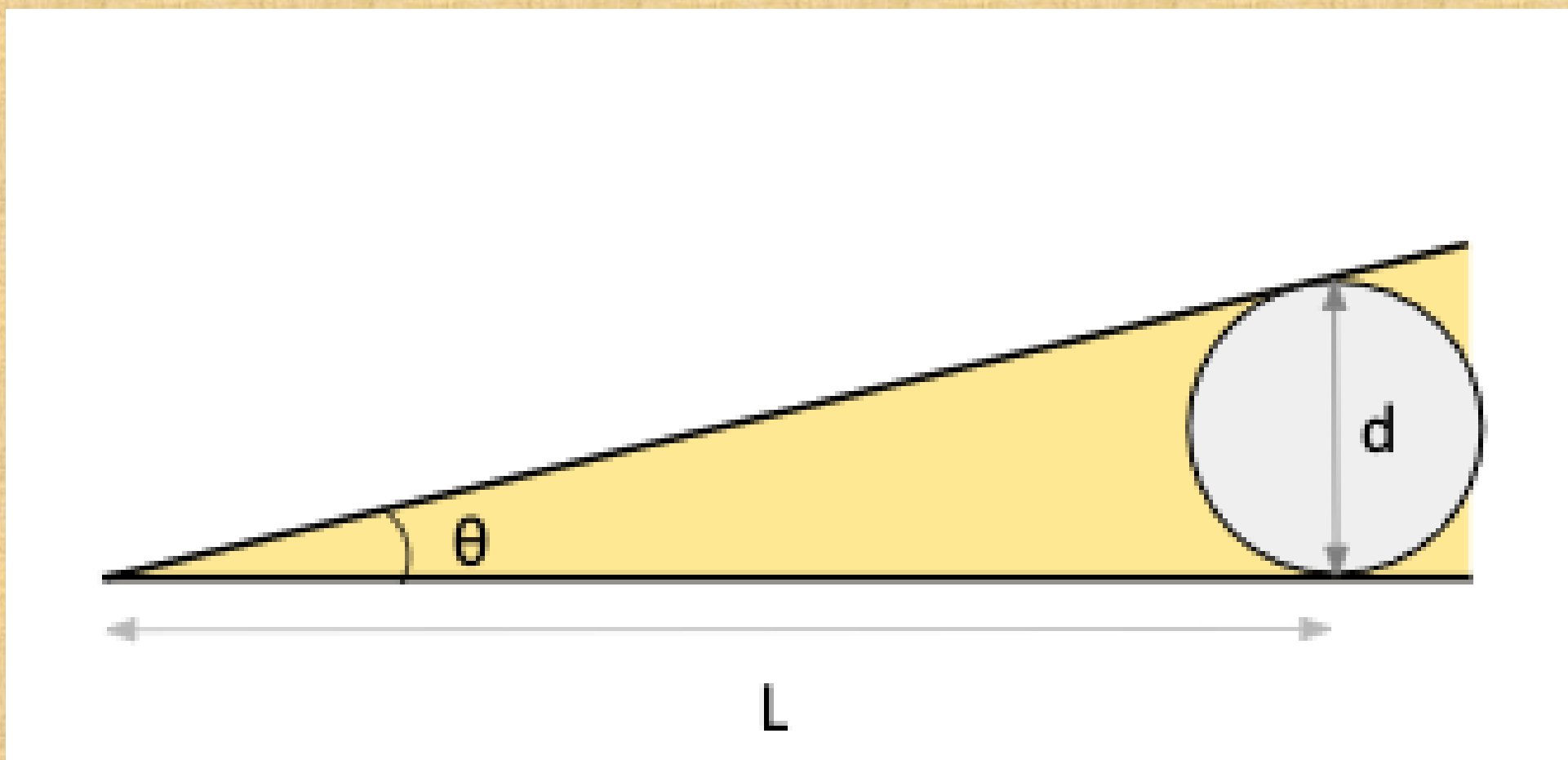
$$\frac{r_n^2}{R} = \frac{n\lambda}{\mu}$$

$$r_n^2 = \frac{nR\lambda}{\mu}$$

If the lens, light source and film medium
remain same, 'R', ' λ ' and ' μ ' remain constant

$$r_n^2 \propto n$$

WEDGE SHAPED THIN FILM: DIAMETER OF A WIRE



$$\tan \theta = \frac{d}{L}$$

As ' θ ' is very small for a wedge-shaped thin film $\tan \theta \sim \theta$

$$\theta = \frac{d}{L}$$

$$\beta = \frac{\lambda}{2\mu\theta}$$

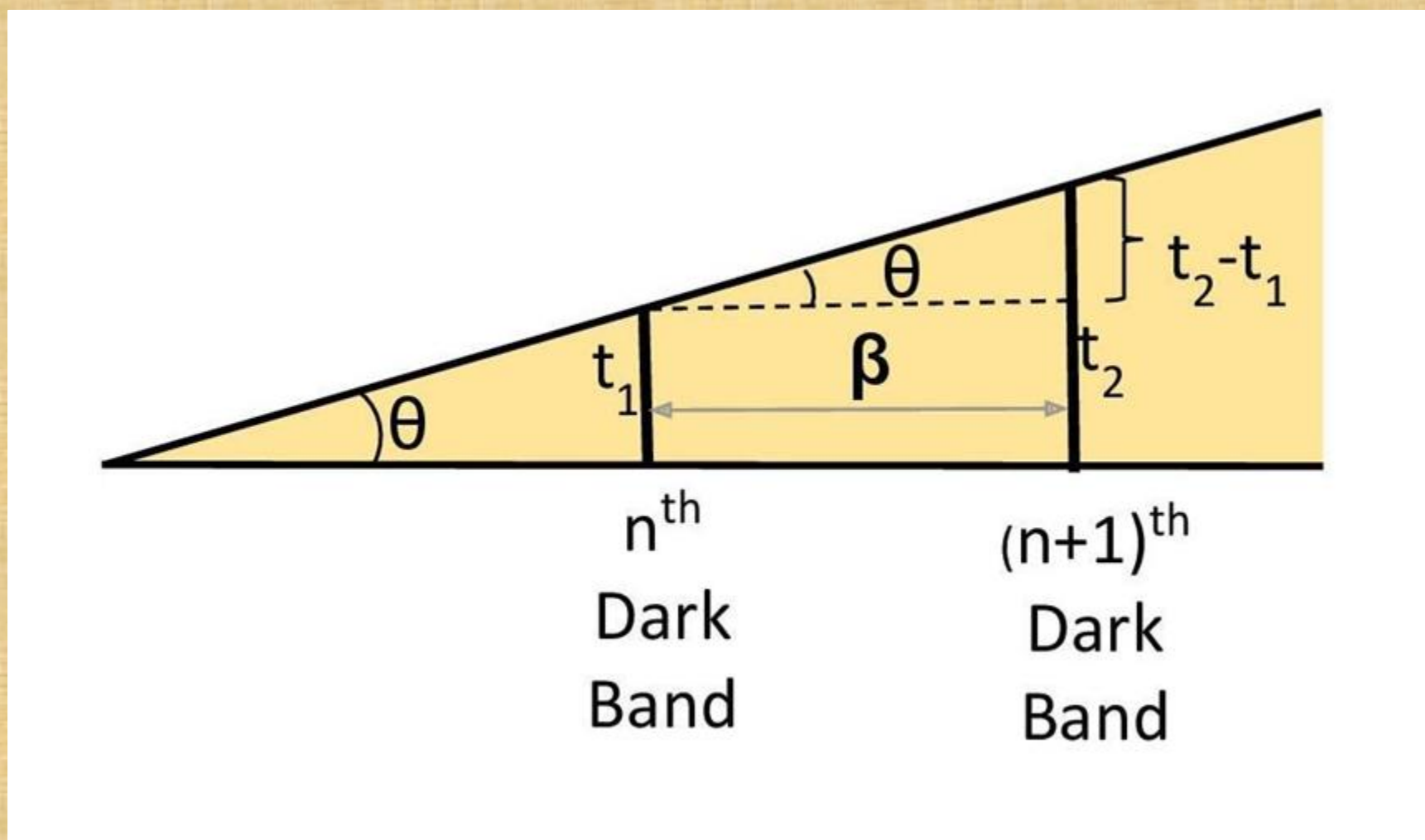
Expression of Fringe width for wedge-shaped thin film

$$\beta = \frac{\lambda L}{2\mu d}$$

Rearranging the equation for ' d '

$$d = \frac{\lambda L}{2\mu\beta}$$

EXPRESSION FOR FRINGE WIDTH DERIVATION



$$2\mu t_1 \cos(r + \theta) = n\lambda$$

Destructive interference condition for the n^{th} dark band

$$2\mu t_2 \cos(r + \theta) = (n + 1)\lambda$$

Destructive interference for the $(n+1)^{\text{th}}$ dark band

$$2\mu t_1 = n\lambda$$

In case of normal incidence, the angle of refraction 'r' tends to zero and as ' θ ' is very small $(r+\theta)$ tends to zero and hence $\cos(r+\theta)=1$

$$2\mu t_2 = (n + 1)\lambda$$

Subtract

$$2\mu(t_2 - t_1) = \lambda$$

$$\tan \theta = \frac{t_2 - t_1}{\beta}$$

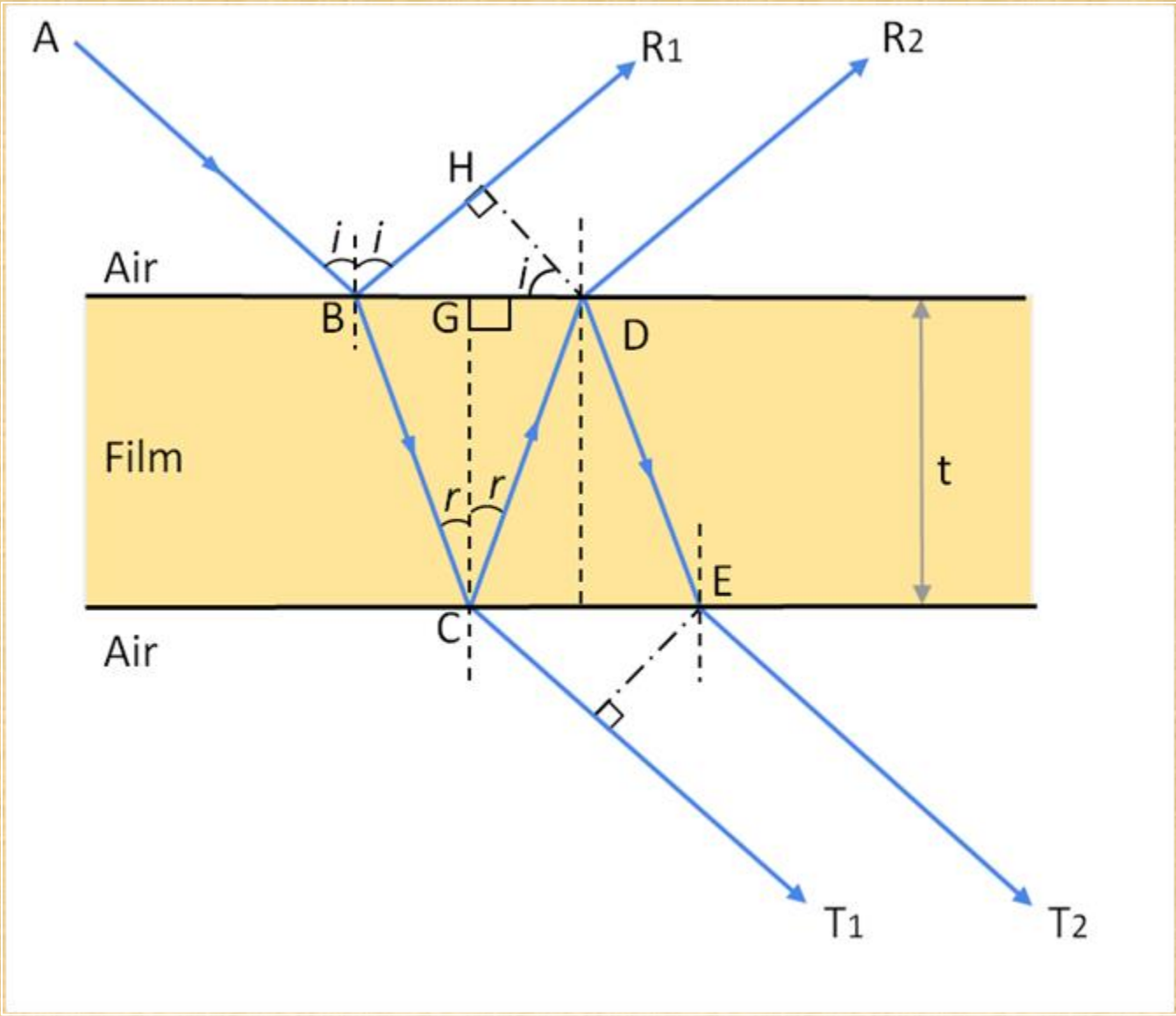
$$\tan \theta = \frac{\lambda}{2\mu\beta}$$

$$\beta = \frac{\lambda}{2\mu \tan \theta}$$

$$\beta = \frac{\lambda}{2\mu\theta}$$

EXPRESSION FOR OPTICAL PATH DIFFERENCE

7 times for 8 marks



$$\Delta OPD = \Delta = (BC + CD) \times \mu - (BH \times 1)$$

$$BC = CD$$

$$\Delta OPD = \Delta = 2\mu BC - BH$$

$$\Delta = \left(\frac{2\mu t}{\cos r}\right) - BH$$

$$\Delta = \left(\frac{2\mu t}{\cos r}\right) - BD \sin i$$

$$BD = 2BG$$

$$\Delta = \left(\frac{2\mu t}{\cos r}\right) - 2BG \sin i$$

$$\Delta = \left(\frac{2\mu t}{\cos r}\right) - 2t \tan i \sin i$$

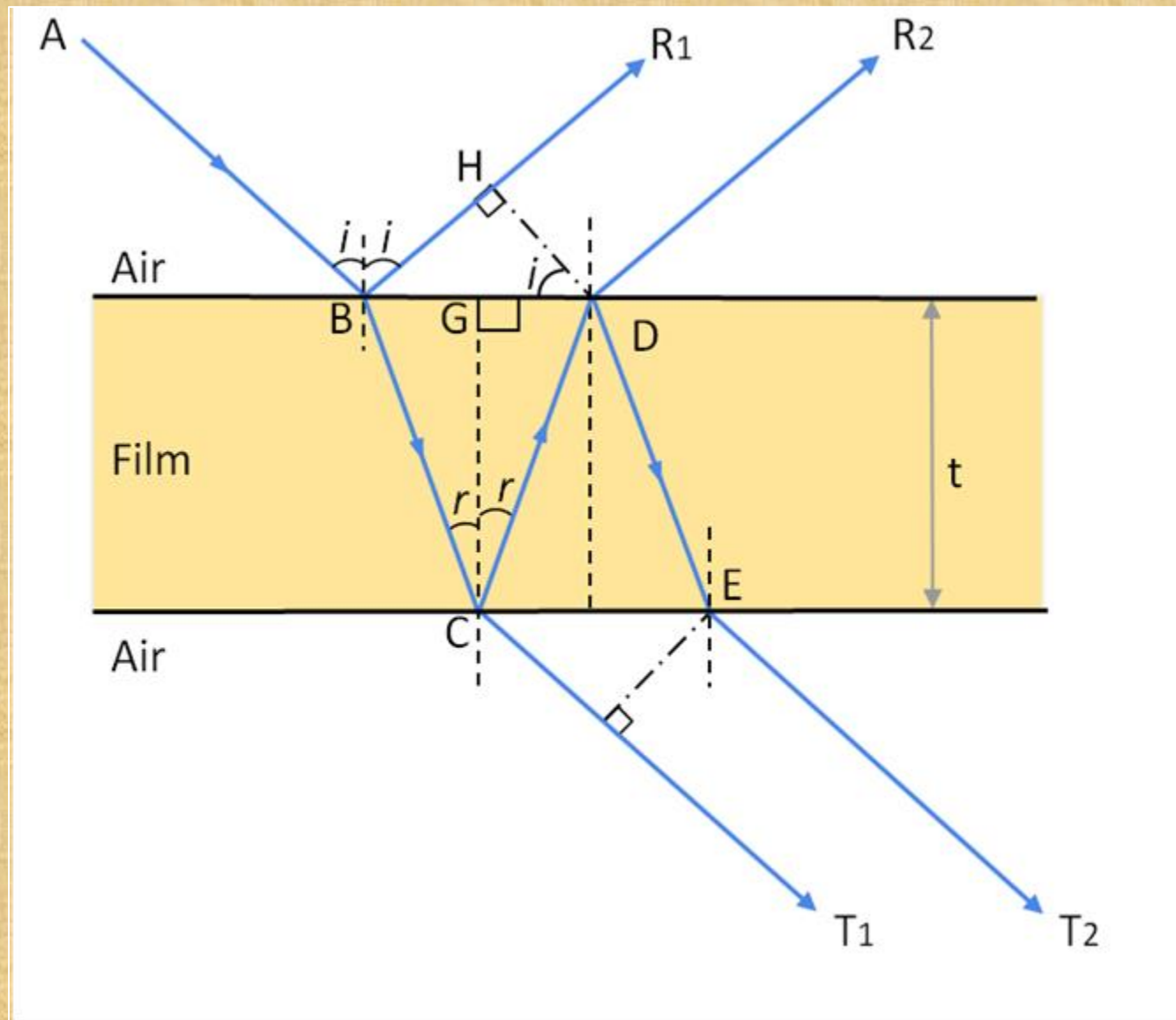
$$\mu = \sin i / \sin r$$

$$\Delta = \left(\frac{2\mu t}{\cos r}\right) - 2\mu t \tan r \sin r$$

$$\Delta = \left(\frac{2\mu t}{\cos r}\right) (1 - \sin^2 r)$$

$$\Delta OPD = 2\mu t \cos r$$

EXPRESSION FOR OPTICAL PATH DIFFERENCE CONTINUATION



$$\Delta OPD = 2\mu t \cos r$$

$$\Delta = 2\mu t \cos r - \frac{\lambda}{2}$$

$$\Delta = n\lambda$$

$$2\mu t \cos r = (2n - 1) \frac{\lambda}{2}$$

$$\Delta = (2n - 1)\lambda$$

$$2\mu t \cos r = n\lambda$$